

Volleyball-Group-Activity-Recognition

Omar Ayman Tawfiq Bakr  

Abstract—A condensed summary of the Volleyball Group Activity Recognition project: the data pipeline, the generic multi-baseline loader, and the four completed baselines — B1, a single-frame fine-tuned ResNet-50 (62.6% test accuracy / 0.630 macro F1); B3, a person-then-group crop classifier (60.7% / 0.589); B4, a temporal classifier that runs an LSTM over B1’s frozen frame features (66.1% / 0.673, best macro F1); and B5, a per-player temporal model that pools LSTM summaries of each player’s crop sequence (66.3% / 0.619, best accuracy). The dataset (Ibrahim et al., CVPR 2016) contains 55 videos / 4,830 clips with 8 scene-level group activities and 9 player-level actions.

Keywords— Volleyball, Group Activity Recognition, CNN, RNN, LSTM, ResNet50, Classification, Multiclass Classification, Transfer Learning

1. Problem & Dataset

Volleyball Group Activity Recognition is a hierarchical classification problem: predict the scene-level group activity (8 classes: 1/r-pass, -spike, -set, -winpoint) and the per-player action (9 classes: blocking, digging, falling, jumping, moving, setting, spiking, standing, waiting). The dataset comes from “A Hierarchical Deep Temporal Model for Group Activity Recognition” (Ibrahim et al., CVPR 2016; official repo, Kaggle mirror).

Key facts: **55 videos, 4,830 clips** (train/val/test = 2,152/1,341/1,337 clips over disjoint video splits of 24/15/16). Each clip has **41 frames**; the clip folder is named after its annotated *middle frame*. Both label levels are heavily skewed: standing is $\approx 68\%$ of person actions, and 1/r_winpoint are $\approx 2.5\times$ rarer than the other group classes. The group classes come in **left/right mirror pairs**, so horizontal geometry is part of the label — the transform pipelines therefore use no horizontal flips and warp inputs directly to 224×224 so the full court (or the full player) always stays in view.

2. Data Pipeline

Three raw annotation sources (action detections, person detections, player tracking) are unified by a two-stage parser (`src/json_parser.py`) into one master JSON keyed `video_id/clip_id`, then enriched with the group-activity label per clip (looked up per video against each clip’s own middle frame) and cached as a pickle (~ 247 MB) for instant metadata loading. Frames are served either from a memory-mapped LMDB (local) or straight from disk (Kaggle); both backends expose the identical `VolleyballDataset`, which covers every baseline through flags: full frames or per-player crops, 1 or 9 frames per clip. A custom collate pads the variable player count per clip and returns a validity mask so padded slots never reach the loss or the pooling.

3. Baseline Roadmap

Table 1 lists the planned progression of models, each adding temporal and/or player-level structure on top of the previous one.

Table 1. Baseline models. All eight complete with results; B8 (team-split pooling) leads at 85.6% test accuracy.

Model	Temporal	Player	Scene
B1	—	—	Image clf.
B3	—	Crop clf.	Concat-pool + MLP
B4	LSTM	—	LSTM
B5	LSTM/player	Concat-pool	MLP
B6	LSTM/image	Max-pool/frame	LSTM + skip Conv1d
B7	LSTM ₁ + LSTM ₂	Pool/frame	LSTM ₂ + skips
B8	LSTM ₁ + LSTM ₂	Team-split	LSTM ₂ + skips

4. Baseline 1: Single-Frame Classifier

Fine-tunes a pretrained ResNet-50 on the clip’s middle frame for the 8 group classes. Two-stage schedule: linear probe (head only, 10 epochs, lr 10^{-3}), then partial fine-tune of `layer3/4` + head (up to 100 epochs, backbone lr 2.5×10^{-4} , head $3\times$, cosine annealing, batch 16, frozen BatchNorm held in eval).

Table 2. Baseline 1 test-set metrics.

Metric	Value
Accuracy	62.60%
Macro F1	0.630
Test Loss	1.42

Fig. 1 shows the per-class confusion on the test split.

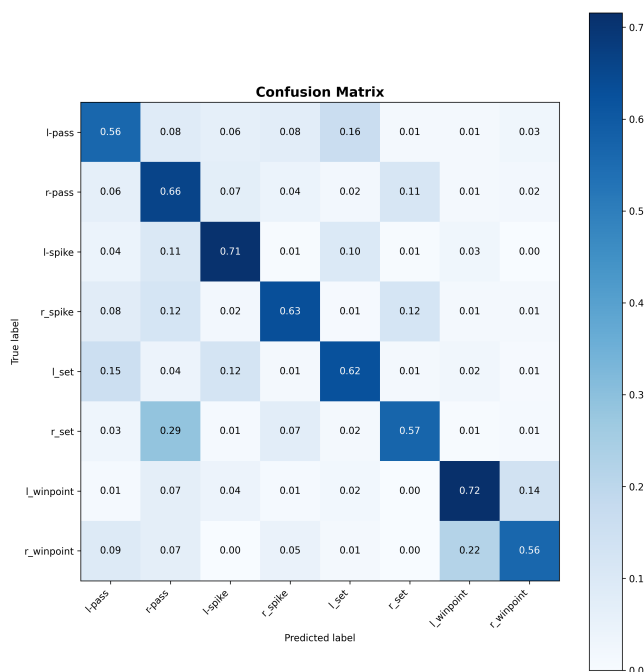


Figure 1. Baseline 1 test confusion matrix.

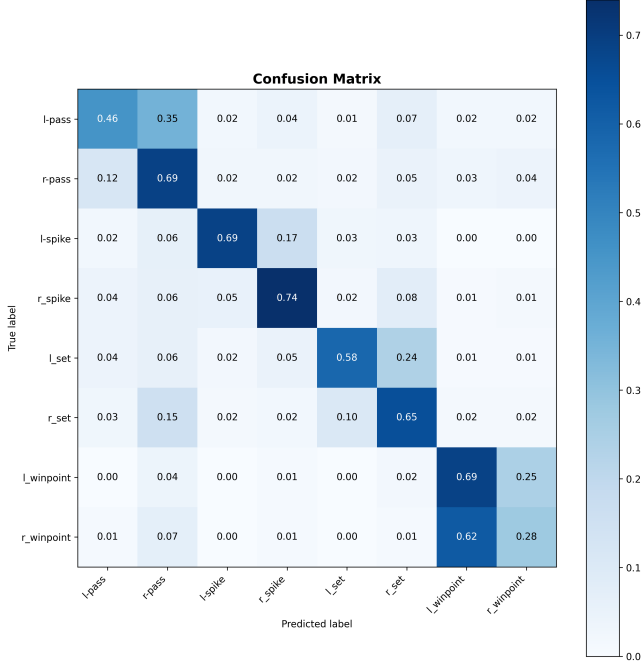
5. Baseline 3: Person-Then-Group

Stage A fine-tunes ResNet-50 on individual player crops for the 9 person actions with inverse-frequency class-weighted CE (without it the loss collapses onto `standing`). **Stage B** freezes that backbone, extracts a ($P, 2048$) feature matrix per clip, concatenates masked `max`- and `mean`-pooling across players (2×2048), and trains an MLP ($4096 \rightarrow 512 \rightarrow 8$, Dropout 0.4) with class-weighted CE for the group classes. SGD (momentum 0.9, Nesterov), 50 epochs and patience 10 per stage, batch 16 clips, `nn.DataParallel` on multi-GPU hosts. Stage A reaches 70.4% validation accuracy (macro F1 0.512) on the person actions; Stage B 60.0% (0.584) on the group activities.

Fig. 2 shows the dominant error mode: left/right mirror confusion within each activity pair, worst on the rare `r_winpoint`. The main open issue is **overfitting** (train $\approx 95\%$ vs. val 58–70%), to be addressed with stronger augmentation, higher dropout, and earlier stopping in a follow-up run.

Table 3. Baseline 3 test-set metrics.

Metric	Value
Accuracy	60.73%
Macro F1	0.589
Test Loss	1.09

**Figure 2.** Baseline 3 test confusion matrix.

6. Baseline 4: Temporal Image Classifier

Each clip’s 9-frame window is pushed through a **frozen** feature extractor loaded from Baseline 1’s fine-tuned backbone, producing a (9, 2048) feature sequence per clip. A single-layer LSTM (hidden 512) consumes the sequence and its final hidden state is classified by a small MLP head (512 → 256 → 8, Dropout 0.3). Only the LSTM + head train (≈ 5.4 M parameters; SGD $1r 10^{-3}$, cosine annealing, class-weighted CE, batch 16 clips).

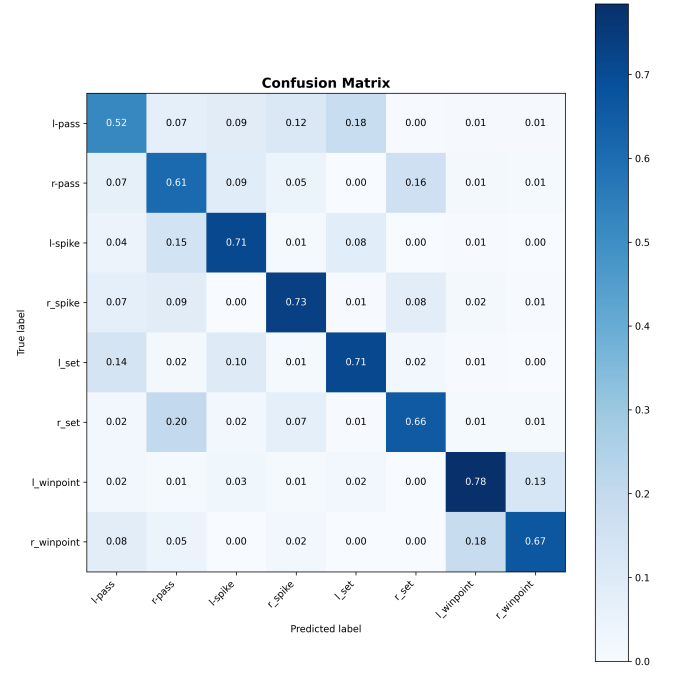
Table 4. Baseline 4 test-set metrics.

Metric	Value
Accuracy	66.12%
Macro F1	0.673
Test Loss	1.05

B4 is the strongest baseline so far: temporal context over B1’s own features is worth **+3.5 accuracy points / +0.043 macro F1** versus B1’s single-frame result — the first direct evidence in this project that motion carries group-activity signal. Convergence is very fast (best validation F1 at epoch 4; training accuracy $\approx 99\%$ by epoch 14, where early stopping fired), repeating the overfitting pattern of B1/B3. Fig. 3 shows the test confusion matrix.

7. Baseline 5: Temporal Person-Level Model

A two-stage temporal extension of B3. **Stage A** feeds each player’s 9-crop sequence through B3’s **frozen** Stage-A person-action backbone and trains a shared LSTM (hidden 3072, 1 layer) + head to classify the 9 person actions from the sequence’s final hidden state. **Stage B** freezes the Stage-A model entirely, computes one LSTM summary

**Figure 3.** Baseline 4 test confusion matrix.

per player per clip, pools the summaries across players with masked concat (max || mean, classifier input 2×3072), and trains an MLP head ($6144 \rightarrow 3072 \rightarrow 1536 \rightarrow 8$, LayerNorm + Dropout 0.2) on the 8 group activities. Training uses SGD ($1r 10^{-3}$, cosine annealing in Stage B), class-weighted CE in both stages, and gradient accumulation (effective batch $8 = \text{micro } 4 \times 2$) since each clip expands to $9 \times \sim 12$ crop images.

Table 5. Baseline 5 test-set metrics.

Metric	Value
Accuracy	66.34%
Macro F1	0.619
Test Loss	0.97

B5 achieves the **best test accuracy and test loss of baselines B1–B5** (66.3% / 0.97 vs. B4’s 66.1% / 1.05) and is markedly stronger on the core activities — spike F1 reaches 0.78–0.80 and set 0.71 on both court sides, with cross-activity confusion nearly eliminated — direct evidence that per-player temporal modeling adds signal beyond B4’s scene-level features. Its macro F1 (0.619) nevertheless trails B4 (0.673) because of a single failure mode: **the l/r_winpoint confusion is the worst left/right confusion of any baseline in this project**. As Fig. 4 shows, r_winpoint collapses to 0.08 recall (F1 0.13), with 83% of true r_winpoint clips predicted as l_winpoint. The cause is structural: orderless max/mean pooling over all ~ 12 players erases which side of the net each player occupies; winpoint clips show both teams in near-identical celebration poses, so side identity is the only discriminative signal — and the pooling destroys it. B8’s team-split pooling (pool each team’s six players separately, then concatenate) directly targets this failure.

8. Baseline 6: Scene-Level Temporal Model

B6 moves the temporal model from the player level (B5) to the **scene level**. Each frame’s player crops pass through B3’s **frozen** Stage-A backbone and are masked max-pooled across players *per frame*, yielding a (9, 2048) scene sequence per clip. An LSTM (hidden 512, 1 layer) consumes it; instead of keeping only the last hidden state, its 9 hidden states are concatenated *along the time axis* with a linear

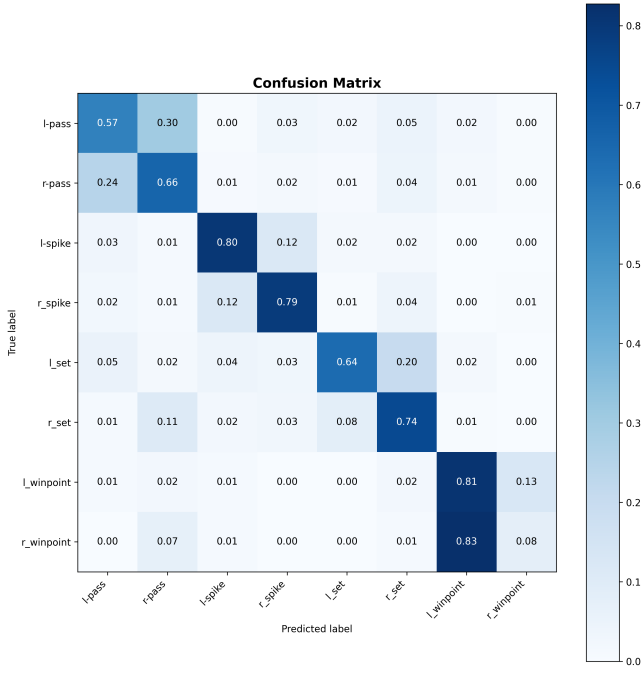


Figure 4. Baseline 5 test confusion matrix. Note the $r_winpoint$ row: 83% of its clips are predicted as $l_winpoint$ — the side-blind pooling failure discussed in the text.

projection of the pooled per-frame features (skip connection), giving (18, 512), which a two-stage Conv1d with a global temporal kernel ($512 \rightarrow 256, k=18$, then $256 \rightarrow 128, k=1$) collapses into a 128-dim clip summary. **Stage A** pretrains LSTM + projection + Conv1d on the 9 person actions using single-player tracks ($P=1$, pooling is an identity). **Stage B** is two-phase: a 10-epoch **linear probe** (temporal model frozen, MLP head only, $lr 10^{-3}$), then a **joint fine-tune** that unfreezes the temporal machinery with differential learning rates (temporal 10^{-4} , head 10^{-3} , cosine annealing, 50 epochs). The ResNet extractor stays frozen throughout; training otherwise matches B5 (class-weighted CE, label smoothing 0.01, effective batch $8 = \text{micro } 4 \times 2$).

Table 6. Baseline 6 test-set metrics.

Metric	Value
Accuracy	70.53%
Macro F1	0.686
Test Loss	1.04

B6 is the first baseline to lead on *both* metrics: accuracy gains **+4.2 points over B5** (70.5% vs. 66.3%) and macro F1 passes B4’s previous best (0.686 vs. 0.673). Scene-level temporal modeling over pooled person features corresponds to the paper’s “two-stage model without LSTM 1” (74.7% there); the remaining ~ 4 -point gap is consistent with our frozen rather than end-to-end backbone. The two-phase Stage B was decisive: the linear probe plateaued at 0.426 validation F1 — a frozen LSTM pretrained only on person actions is a poor group-activity descriptor — while joint fine-tuning lifted it to 0.646 (+0.22). The scene LSTM must see the group labels; the probe merely gives the head a stable start before the temporal weights move.

Per class (Fig. 5), B6 is strongest on the acting-player activities (spike F1 0.86/0.82, set 0.72/0.69) and **halves B5’s winpoint collapse without any team-aware pooling**: $r_winpoint$ recall recovers from 0.08 to 0.47 (F1 0.13 $\rightarrow$ 0.49) and $l_winpoint$ reaches F1 0.55 — scene-level temporal context evidently carries side information that per-player last-hidden summaries lost. Left/right confusion is nevertheless now the dominant remaining error: 47% of

true $r_winpoint$ is predicted $l_winpoint$, l/r-pass leak 20–22% into each other, and 21% of l_set is called r_set — the model is usually right about *what* happened but guesses *which side*. All-player max pooling remains side-blind; B8’s team-split pooling targets exactly this. Overfitting persists (Stage A train F1 0.96 vs. val 0.54; fine-tune 0.93 vs. 0.64).

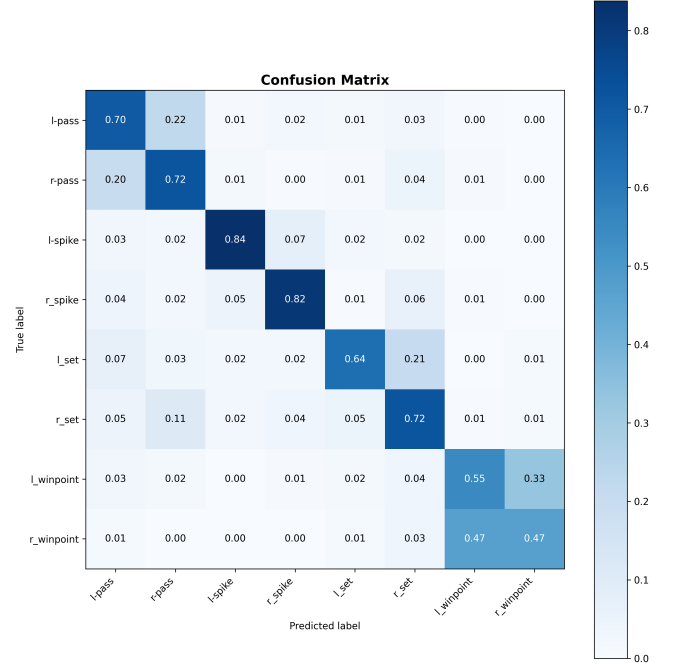


Figure 5. Baseline 6 test confusion matrix. Winpoint recovers from B5’s collapse ($r_winpoint$ recall 0.08 $\rightarrow$ 0.47), but left/right leakage — winpoint, pass, and set — is now the dominant error mode.

9. Baseline 7: Full Hierarchical Model

B7 is the paper’s two-stage hierarchical model with skip connections at both levels. A player-level **LSTM₁** runs over each player’s 9-frame track; players are pooled per frame into a scene sequence that a scene-level **LSTM₂** consumes. Two skips are added on top of the paper: (1) *player-level, feature-axis* — each timestep’s per-player vector is [LSTM₁ output || Linear-projected backbone features] (the paper’s $fc7 \parallel \text{hidden}$ trick), keeping appearance beside the temporal summary while the time axis survives for LSTM₂; (2) *scene-level, time-axis* — B6’s recipe of concatenating LSTM₂’s hidden states with a projection of the pooled scene sequence along time, then a global-kernel Conv1d fusion. Stage A pretrains LSTM₁ + projection on the 9 person actions ($P=1$ tracks); Stage B is two-phase (probe with the player model frozen, then joint fine-tune unfreezing LSTM₁), selecting on validation accuracy. Optimizer is AdamW.

Table 7. Baseline 7 test-set metrics (run 3).

Metric	Value
Accuracy	73.75%
Macro F1	0.701
Test Loss	0.89

B7 is the **strongest model without team-split pooling** (only B8 beats it): accuracy +3.2 points over B6 (73.8% vs. 70.5%), macro F1 0.701 vs. 0.686, and a lower test loss (0.89). The full hierarchy pays off — an adapting player-level LSTM feeding a scene-level LSTM beats B6’s single scene-level LSTM over pooled frames. The result hinges on the **two-phase Stage B**: with LSTM₁ frozen the probe plateaus at

~0.51 validation accuracy, and unfreezing it for the joint fine-tune lifts validation accuracy to ~0.68 (+0.17). This is the same lesson as B6 (pretrained temporal weights must adapt to the group task) and explains why B7’s earlier *frozen* single-phase run scored only 65.8%, below B6 — unfreezing turned that into +7.9 points. What B7 still does not fix is the left/right confusion: it pools all players side-blind, so winpoint/pass/set leak across sides. That is exactly B8’s target.

10. Baseline 8: Team-Split Pooling

B8 makes a single architectural change to B7 — the paper’s group-style pooling. Instead of pooling all ~12 players into one scene vector (side-blind), each team’s players are pooled *separately* and the two team vectors are concatenated, so the scene vector keeps *which side did what* — the signal every earlier baseline erases, and the direct fix for the dominant left/right confusion (winpoint, pass, set). Both of B7’s skips are kept unchanged; only the scene vector doubles in width (LSTM₂ input becomes $4H_1$ for max/mean or $8H_1$ for concat). Team membership comes from the data loader in team mode (`with_teams=True`), which emits a per-player `team_ids` tensor (0/1 by court side, -1 for padding) derived once per clip from box center-*x* ordering — the paper’s split, applied as a fixed per-track label so it coexists with the track-ID ordering the LSTMs require. Uneven or empty teams are handled by masking (a missing team pools to a zero vector).

Table 8. Baseline 8 test-set metrics (run 1).

Metric	Value
Accuracy	85.64%
Macro F1	0.855
Test Loss	0.51

B8 is the **best model in the project by a wide margin**: +11.9 accuracy and +0.154 macro F1 over B7 (85.6% vs. 73.8%), clearing the CVPR-2016 full model (81.9%). The single change from B7 — pooling each team separately — is responsible, and its effect is visible exactly where predicted. As Fig. 6 shows, the **left/right confusion is fixed**: `r_winpoint`, which collapsed to ~0.08 recall in B5 and ~0.38 in B7 (47% predicted as `l_winpoint`), now sits at **0.86** with 5% cross-side leak, and `l_winpoint` reaches 0.83; the l/r-pass swap (20–24% in B6/B7) is gone (both 0.86–0.87); spike is near-perfect (0.93/0.90). What remains is *within-side* confusion (e.g. `l_set`→`l_spike` 12%), not left/right — the model now reliably knows which team acted. Team pooling is strong even without fine-tuning: the probe alone (LSTM₁ frozen) reached ~0.83 validation accuracy, already above B7’s fully-fine-tuned 0.68.

One caveat on the ablation: B8 used the `baseline3_stage_a_run3` backbone while B7 used `run2` (person-action val acc 75.0% vs. 70.4%), so the +11.9 gain conflates team-split pooling with a slightly stronger backbone. The pooling is clearly dominant — the paper’s own team-pooling ablation is +11.6, and the probe-alone result isolates the effect — but a perfectly clean comparison would re-run B7 on `run3`.

11. Next Steps

(i) **Clean the B7/B8 ablation** by re-running B7 on the `run3` backbone, so the +11.9 jump is attributed to team-split pooling alone. (ii) **Reduce overfitting** across all baselines — stronger augmentation, higher dropout, and earlier stopping remain the levers (B8’s fine-tune reaches ~97% train accuracy against ~84% validation). (iii) Beyond the ladder, the extension paper’s graph/relational models are the natural next step for modeling inter-player structure explicitly rather than via pooling.

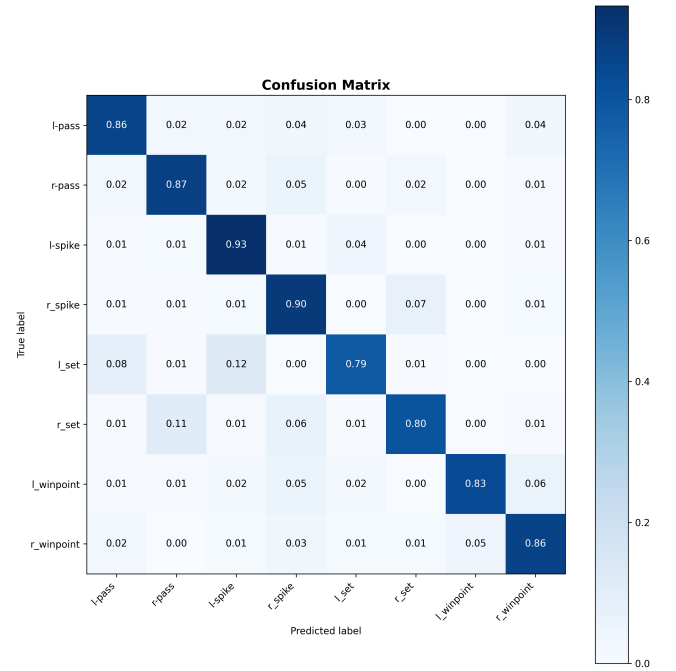


Figure 6. Baseline 8 test confusion matrix. Team-split pooling eliminates the left/right leakage: winpoint recovers to 0.83/0.86 on the diagonal (from B7’s ~0.38), and pass/spike/set no longer swap sides. Remaining errors are within-side (e.g. set vs. spike).